

1 **ONLINE SUPPLEMENT**

2 **Intensive Blood Pressure Lowering Improves Left Ventricular Hypertrophy in Older**
3 **Patients with Hypertension: The Strategy of Blood Pressure Intervention in the Elderly**
4 **Hypertensive Patients (STEP) Trial**

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METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Study Population and Design

The design and results of the STEP trial have been reported previously(1, 2). Briefly, STEP was a randomized, controlled, open-label trial that were conducted at 42 clinical centres throughout China and compared the clinical outcomes of intensive treatment (SBP target, 110 to <130 mmHg) versus those of standard treatment (130 to <150 mmHg) in a large sample of older patients with hypertension. The study inclusion criteria were as follows: age 60–80 years; Han ethnicity; SBP 140–190 mmHg or on antihypertensive medication. The exclusion criteria were previous stroke, mental impairment, uncontrolled diabetes, or a serious life-limiting condition. Patients with both qualified ECG images at baseline and during follow-up were included in current analysis (Figure S1). The trial was approved by the ethics committee of FuWai Hospital and each clinical site and conducted in accordance with the principles outlined in the Declaration of Helsinki, and all patients provided their written informed consents.

Ascertainment of LVH

A standard 12-lead electrocardiogram (ECG) was obtained at baseline, year 3, and year 4, the follow-up ECG was defined as the last available one for each patient. The ECGs were obtained in a consistent fashion (calibration, 10 mm/mV; speed, 25 mm/s) according to the standard operating procedure used at all study sites. All ECGs were interpreted centrally at the STEP ECG reading centre. All ECG tracings were initially screened visually for technical errors and inadequate quality before being viewed by two experienced observers blinded to patient characteristics and treatment allocation. The patient selection process is shown in Figure S1. Interobserver and intraobserver agreement was examined using a sample of ECGs with varying image quality and found to be excellent (intraclass correlation coefficient > 0.8 for all measurements).

LVH was defined by sex-specific Peguero-Lo Presti criteria, computed as summing the deepest S wave amplitude of any lead (S_D) and the S wave amplitude of lead V_4 (SV_4) using the PR segment as the baseline with a cut-off point of $\geq 2,300 \mu V$ for women and $\geq 2,800 \mu V$ for men(3). LVH was considered present or absent (or changed from one to the other) according to crossing of these cut-off points up or down. In addition to using LVH as a categorical/binary variable, the sum of S_D and SV_4 was also examined as a continuous variable (referred to from now onwards as the Peguero-Lo Presti index).

Intervention, measurements, and follow-up

Demographic data were collected at baseline before randomization. All patients received antihypertensive medication, including olmesartan (an angiotensin II receptor blocker), amlodipine (a calcium channel blocker), and, if needed, hydrochlorothiazide (a diuretic). Office BP was measured by a trained trial staff member using the same validated office BP monitor (Omron Healthcare Group, Kyoto, Japan). Sitting brachial BP was measured in the upper right arm using an appropriately sized cuff after 5 min of rest and calculated as the average of three readings obtained at 1-minute intervals. Further details of the assessment of BP, adjustment of medication, and the antihypertensive drug regimens used throughout the trial are provided elsewhere(1).

Patients were scheduled for follow-up visits at 1, 2, and 3 months and every 3 months thereafter until the end of the study. At each visit, a structured interview was performed regardless of treatment group to obtain self-reported information on cardiovascular outcomes. The primary outcome was defined as a composite of stroke, acute coronary syndrome (myocardial infarction or hospitalization for unstable angina), hospitalization for acute decompensated heart failure, atrial fibrillation, and cardiovascular death. All events were evaluated by an independent endpoint adjudication committee blinded to treatment allocation.

Statistical Analysis

Continuous variables were compared between groups using two sample *t*-tests if normally distributed and the Wilcoxon rank sum test if non-normally distributed. Categorical variables were compared using the likelihood ratio, chi-squared test, or Fisher's exact test. Cox proportional hazards regression models with stratification of clinical sites and adjustment of potential covariables (including age, sex, body mass index, baseline SBP, diabetes mellitus, chronic heart disease, baseline total cholesterol, uric acid, baseline estimated glomerular filtration rate, smoking status, alcohol consumption, aspirin use and statin use) were used to compare the time to first detection of LVH in patients without LVH at baseline and the time to first evidence of regression of LVH in patients with LVH at baseline. Follow-up was censored on the date of the last ECG. Interactions between treatment effect and our pre-specified subgroups, namely, age (<70 years vs. ≥70 years), sex, SBP tertile (≤138, >139 to <151, and ≥152 mmHg), and diabetes status, were assessed using a likelihood ratio test for interaction. To examine whether the impact of intensive treatment on the primary outcome could be explained by its impact on LVH, we examined the magnitude of attenuation of the association between intensive treatment and standard treatment with the primary outcome of STEP after adjusting for LVH or the Peguero-Lo Presti index value as a time-varying covariate. The mediation proportions were calculated using Cox proportional hazards regression models(4).

We performed a sensitivity analysis in which we excluded 430 patients with major intraventricular conduction delay as a result of complete left or right bundle branch block, Wolf-Parkinson-White syndrome, placement of a ventricular pacemaker, and major non-specific conduction delay (all with a QRS duration ≥120 ms) considering that evidence for an ECG diagnosis of LVH in those individuals remains debatable(5).

All statistical analyses were performed using R software version 3.6.3 (R Foundation for Statistical Computing, Vienna, Austria). A two-sided *P*-value of <0.05 was considered statistically significant.

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133 **Table S1. Detailed baseline characteristics of patients enrolled in analysis according to**
134 **treatment groups**

Characteristics	Intensive treatment (N = 3578)	Standard treatment (N = 3563)	P-value
Age, years	66.1 ± 4.8	66.2 ± 4.8	0.286
Age ≥70 yr, no. (%)	842 (23.5)	849 (23.8)	0.769
Male, no. (%)	1670 (46.7)	1640 (46.0)	0.601
Body mass index, kg/m ²	25.6 ± 3.1	25.7 ± 3.2	0.434
Baseline blood pressure, mmHg			
Systolic	146.5 ± 16.7	146.3 ± 16.7	0.713
Diastolic	82.8 ± 10.7	82.4 ± 10.6	0.123
Distribution of systolic blood pressure, no. (%) [*]			0.725
≤138 mmHg	1160 (32.4)	1183 (33.2)	
139-151 mmHg	1161 (32.4)	1156 (32.4)	
≥152 mmHg	1257 (35.1)	1224 (34.3)	
Fasting serum glucose, mmol/L	6.1 ± 1.6	6.2 ± 1.6	0.012
eGFR < 60 ml/min/1.73m ² , no. (%)	51 (1.4)	59 (1.6)	0.429
Lipid profile, mmol/L			
Total cholesterol	4.9 ± 1.1	4.9 ± 1.1	0.726
Triglycerides (IQR)	1.3 (1.0, 1.9)	1.3 (1.0, 1.9)	0.930
High-density lipoprotein cholesterol	1.3 ± 0.3	1.3 ± 0.3	0.883
Low-density lipoprotein cholesterol	2.7 ± 0.9	2.7 ± 0.9	0.756
Uric acid, μmol/L	346.0 ± 88.6	345.6 ± 88.8	0.863
Educational level, no.(%)			0.880
No formal school	178 (5.0)	163 (4.6)	
Primary school	630 (17.6)	645 (18.1)	
Middle or high school	2381 (66.6)	2386 (67.0)	
University or above	388 (10.8)	367 (10.3)	
Smoking status, no. (%)			0.998
Never	2580 (72.1)	2570 (72.1)	
Former	427 (11.9)	426 (12.0)	
Current	571 (16.0)	567 (15.9)	
Alcohol consumption, no.(%)			0.736
Never	2464 (68.9)	2439 (68.5)	
Former	186 (5.2)	200 (5.6)	
Current	928 (25.9)	924 (25.9)	

To be continued

Characteristics		Intensive treatment (N = 3578)	Standard treatment (N = 3563)	P-value
Medical history, no. (%)				
	Diabetes mellitus	684 (19.1)	697 (19.6)	0.655
	Hyperlipidaemia	1329 (37.1)	1303 (36.6)	0.633
	Cardiovascular diseases	224 (6.3)	233 (6.5)	0.426
Antihypertensive agents use, no. (%)		3411 (95.3)	3393 (95.2)	0.880
Statin use, no. (%)		697 (19.5)	691 (19.4)	0.950
Aspirin use, no. (%)		376 (10.5)	366 (10.3)	0.773
The 10-year Framingham risk score $\geq 15\%^{\dagger}$, no. (%)		2318/3564 (65.0)	2275/3550 (64.1)	0.400
Baseline ECG				
	$S_D^{\ddagger}$, μV	1205.9 ± 448.6	1204.8 ± 439.2	0.920
	SV_4 , μV	604.9 ± 401.9	603.1 ± 387.5	0.851
	Peguero-Lo Presti index ($S_D + SV_4$), μV	1810.7 ± 763.8	1807.9 ± 744.2	0.875
	Peguero-Lo Presti-LVH § , no. (%)	593 (16.6)	588 (16.5)	0.936

135 Values are presented as the mean $\pm$ standard deviation, the median (interquartile range), or number
136 (percentage) as appropriate.

137 * The distribution of SBP is presented as the tertile of SBP at baseline.

138 † A 10-year Framingham risk score of $\geq 15\%$ indicates high cardiovascular risk.

139 ‡ S_D was defined as the amplitude of the deepest S wave in any lead.

140 § Peguero-Lo Presti-LVH was ascertained using the following sex-specific cut-off points: $\geq 2,300$

141 μV in women and $\geq 2800 \mu V$ in men. Abbreviations: ECG, electrocardiogram; eGFR, estimated

142 glomerular filtration rate; IQR, interquartile range; LVH, left ventricular hypertrophy; SBP,

143 systolic blood pressure

144 **Table S2. Detailed baseline characteristics of patients enrolled in analysis according to**
145 **baseline LVH**

Characteristics	Baseline Non-LVH (N = 5960)	Baseline LVH (N = 1181)	P-value
Age, years	66.2 ± 4.7	66.1 ± 4.9	0.456
Age ≥70 yr, no. (%)	1421 (23.8)	270 (22.8)	0.094
Male, no. (%)	2766 (46.4)	544 (46.0)	0.827
Body mass index, kg/m ²	25.6 ± 3.14	25.7 ± 3.18	0.081
Baseline blood pressure, mmHg			
Systolic	145.9 ± 16.6	149.1 ± 17.0	0.000
Diastolic	82.4 ± 10.5	83.8 ± 11.0	0.000
Distribution of systolic blood pressure, no. (%) [*]			0.000
≤138 mmHg	2018 (33.8)	324 (27.4)	
139-151 mmHg	1942 (32.5)	375 (31.7)	
≥152 mmHg	1999 (33.5)	482 (40.8)	
Fasting serum glucose, mmol/L	6.1 ± 1.5	6.3 ± 1.8	0.000
eGFR < 60 ml/min/1.73m ² , no. (%)	90 (1.5)	20 (1.6)	0.640
Lipid profile, mmol/L			
Total cholesterol	4.9 ± 1.1	4.9 ± 1.1	0.979
Triglycerides (IQR)	1.4 (1.0, 2.0)	1.3 (1.0, 1.9)	0.122
High-density lipoprotein cholesterol	1.3 ± 0.3	1.3 ± 0.3	0.549
Low-density lipoprotein cholesterol	2.7 ± 0.9	2.7 ± 0.9	0.461
Uric acid, μmol/L	344.5 ± 87.4	352.4 ± 94.5	0.863
Educational level, no.(%)			0.592
No formal school	285 (4.8)	56 (4.7)	
Primary school	1047 (17.6)	228 (19.3)	
Middle or high school	4001 (67.2)	766 (64.9)	
University or above	624 (10.5)	131 (11.1)	
Smoking status, no. (%)			0.132
Never	4291 (72.0)	859 (72.7)	
Former	731 (12.3)	122 (10.3)	
Current	938 (15.7)	200 (16.9)	
Alcohol consumption, no.(%)			0.508
Never	4109 (68.9)	794 (67.2)	
Former	320 (5.4)	66 (5.6)	
Current	1531 (25.7)	321 (27.2)	

To be continued

Characteristics	Baseline Non-LVH (N = 5960)	Baseline LVH (N = 1181)	P-value
Medical history, no. (%)			
Diabetes mellitus	1141 (19.1)	242 (20.4)	0.285
Hyperlipidaemia	2190 (36.7)	442 (37.4)	0.658
Cardiovascular diseases	375 (6.2)	82 (6.9)	0.403
Antihypertensive agents use, no. (%)	5684 (95.4)	1120 (94.8)	0.880
Statin use, no. (%)	1182 (19.8)	206 (17.4)	0.950
Aspirin use, no. (%)	638 (10.7)	104 (8.8)	0.057
The 10-year Framingham risk score $\geq 15\%^{\dagger}$, no. (%)	3807/5937 (64.1)	786/1177 (66.8)	0.064
Baseline ECG			
$S_D^{\ddagger}$, μV	1074.7 $\pm$ 320.0	1864.7 $\pm$ 439.2	0.000
SV_4 , μV	490.7 $\pm$ 286.9	1175.8 $\pm$ 367.4	0.000
Peguero-index ($S_D + SV_4$), μV	1565.3 $\pm$ 504.2	3040.6 $\pm$ 582.0	0.000

146 Values are presented as the mean $\pm$ standard deviation, the median (interquartile range), or number
147 (percentage) as appropriate.

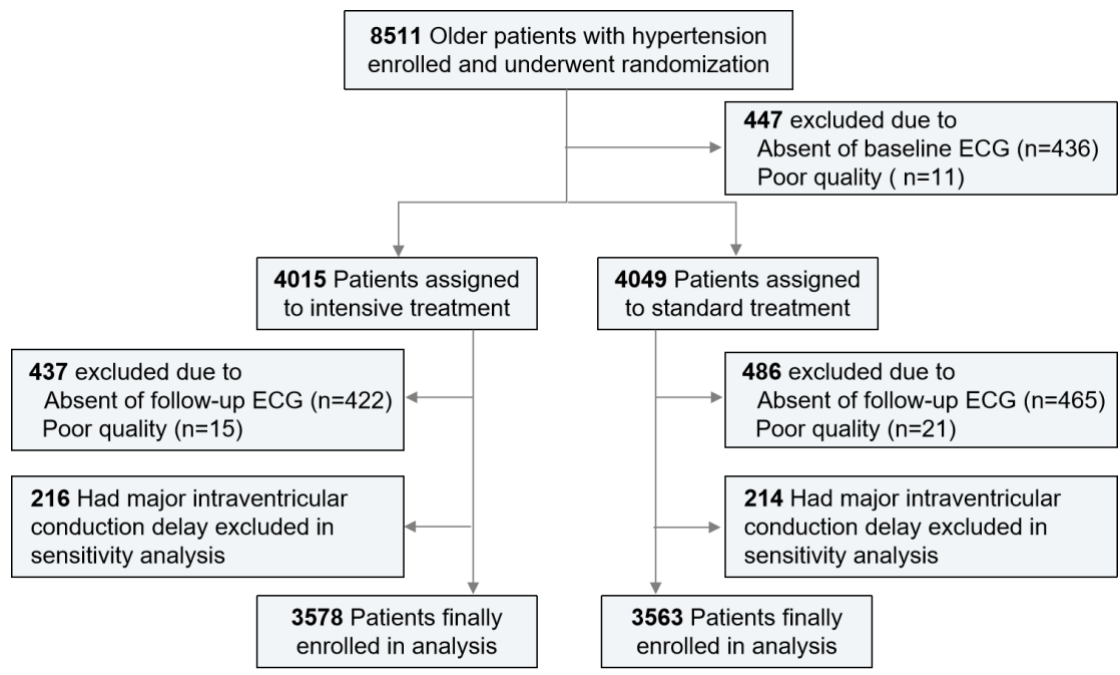
148 *The distribution of SBP is presented as the tertile of SBP at baseline.

149 † A 10-year Framingham risk score of $\geq 15\%$ indicates high cardiovascular risk.

150 $^{\ddagger}S_D$ was defined as the amplitude of the deepest S wave in any lead.

151 Abbreviations: ECG, electrocardiogram; eGFR, estimated glomerular filtration rate; IQR,
152 interquartile range; LVH, left ventricular hypertrophy; SBP, systolic blood pressure

Figure S1. Flow chart showing the patient selection process.



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Major intraventricular conduction delay included all patients with major non-specific conduction delay, complete left and right bundle block, Wolf-Parkinson-White syndrome, and a ventricular pacemaker (all with a QRS duration ≥ 120 ms). CI, confidence interval; ECG, electrocardiogram.

157 **Figure S2. Mean office systolic blood pressure throughout follow-up**

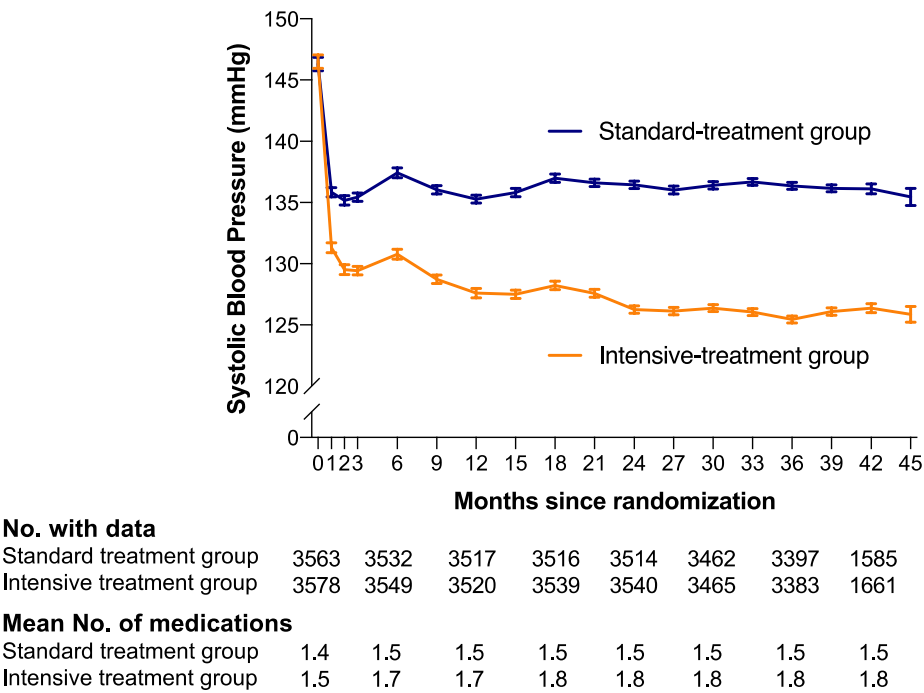
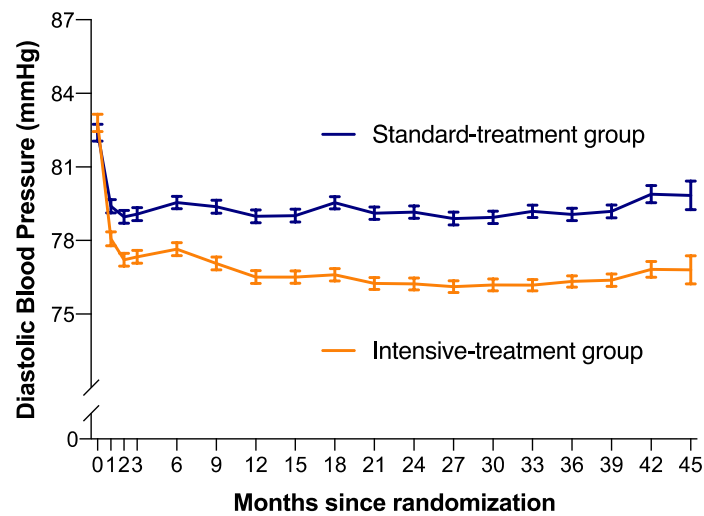
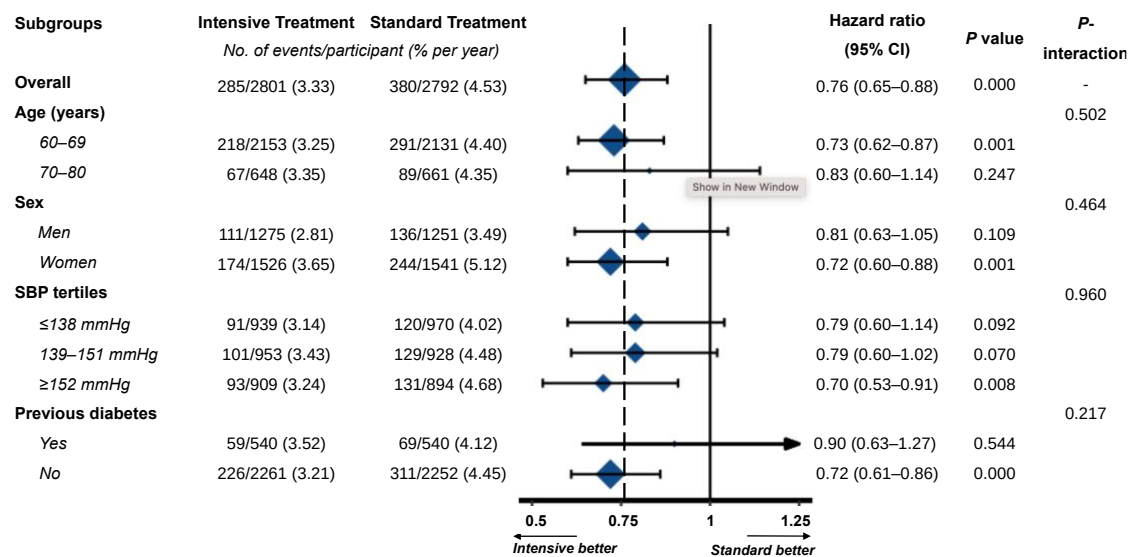


Figure S3. Mean office diastolic blood pressure throughout follow-up



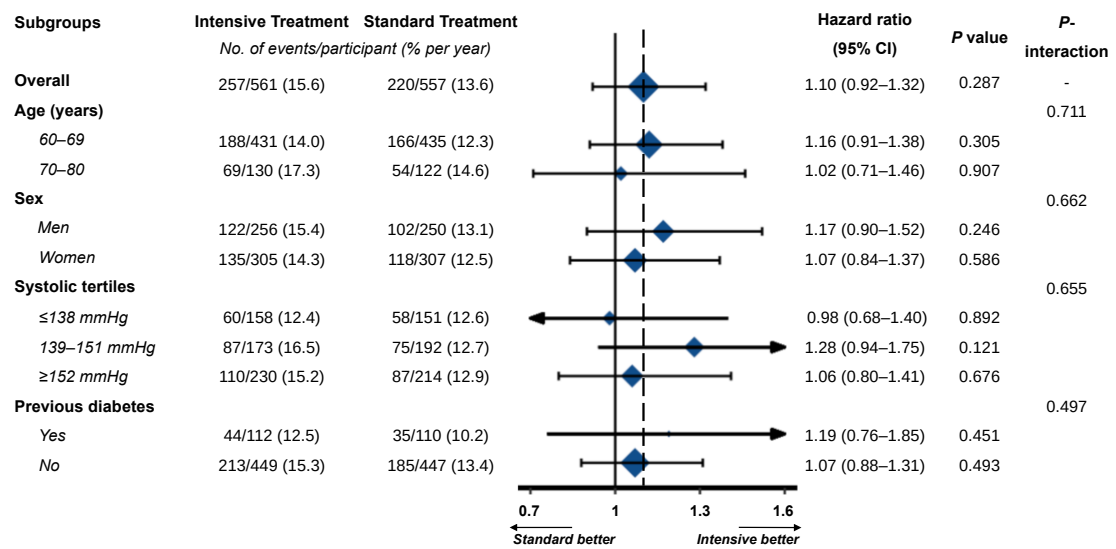
No. with data								
Standard treatment group	3563	3532	3517	3516	3514	3462	3397	1585
Intensive treatment group	3578	3549	3520	3539	3540	3465	3383	1661
Mean No. of medications								
Standard treatment group	1.4	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Intensive treatment group	1.5	1.7	1.7	1.8	1.8	1.8	1.8	1.8

Figure S4. Forest plot of the pre-specified subgroup analysis of the effect of intensive vs. standard treatment on the risk of newly developed LVH throughout follow-up in patients without major intraventricular conduction delay



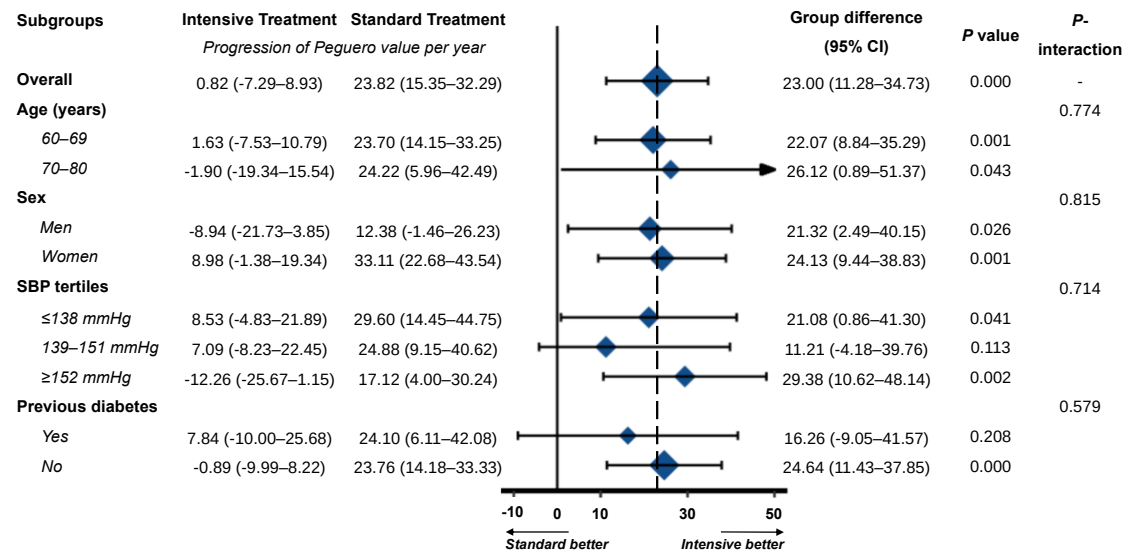
Results are presented as the hazard ratio (HR) and 95% confidence interval (CI). The dashed line represents the overall HR.

Figure S5. Forest plot of the pre-specified subgroup analysis of the effect of intensive vs. standard treatment on the risk of regression of existing LVH throughout follow-up in patients without major intraventricular conduction delay



Results are presented as the hazard ratio (HR) and 95% confidence interval (CI). The dashed line represents the overall HR.

Figure S6. Forest plot of the progression of Peguero-Lo Presti value of intensive vs. standard treatment throughout follow-up in patients without major intraventricular conduction delay



Results are presented as the mean and 95% confidence interval (CI) of between-group difference of the progression Peguero-Lo Presti value per year (μV).